Akan Selim

Distribution Steering • Robust System Design and Optimization • Stochastic Optimal Control • Nonlinear Programming • Astrodynamics

Atlanta, GA • (+1) 470 730 8674 • aselim8@gatech.edu • ResearchGate

Blackbird is still singing the tale of the mighty winter forest...

Aerospace PhD researcher at Georgia Tech and former **Senior GN&C Engineer at ROKETSAN**. My technical expertise comes from astrodynamics, applied trajectory optimization/optimal control for launch vehicles and spacecrafts in formation, stochastic controllers, and theoretical work in **optimal transport, stochastic optimal control and stopping, nonlinear programming, probability measures over probability measures, and contraction metrics**. FieldAI gives me practical exposure to AI, VLAs, large behavior models, Linux/Git/Docker workflows, and validation/verification. I aim to use control, optimization, and measure-valued probability tools for robotics autonomy, computational neuroscience, and understanding learning mechanisms in LLMs/LBMs to push the frontiers in science!

CORE EXPERTISE

Optimal Transport Theory · **Stochastic Calculus** · **Diffusion Models** · **Astrodynamics** · **Trajectory Optimization**

Stochastic/Robust/Optimal/Adaptive Control · **MPC/MPPI** · **Path Integral Control** · **Learning-based Optimal Control**

Sparse Nonlinear Programming / Convex Optimization · **Aerospace / Launch Vehicle System Design** · **Sparse/Hybrid/Multi-disciplinary Optimization**

Data-Driven Dynamic Learning · **Validation/Verification**

Linux · **Git/GitHub** · **Docker** · **Python/PyTorch**

EDUCATION

Georgia Institute of Technology — 2024–Exp. 2029
PhD, Aerospace Engineering; Minor in Mathematics; **GPA: 4.0/4.0**.
Courses: Stochastic Calculus I–II, Optimal Transport Theory, Stochastic Optimal Control, Advanced Nonlinear Control, High-Dimensional Statistics, Hypersonic System Design...

Istanbul Technical University — 2020–2022
MSc, Aeronautics and Astronautics Engineering; **GPA: 3.82/4.0**.
Thesis: Robust Re-entry Trajectory Optimization via Stochastic-Collocation based Ensemble Pseudospectral Optimal Control.
Courses: Adaptive Control Theory, Deep Reinforcement Learning, Learning-based Optimal Control, Optimization Theory...

Istanbul Technical University — 2016–2020
BSc, Astronautical Engineering, Summa Cum Laude.

SKILLS

Programming: Python, PyTorch, JAX, NumPy/SciPy, MATLAB, Octave, Simulink, C...
Tools: Linux, Git/GitHub, Docker, pytest, CVX, IPOPT...
Math/ML: optimal transport, diffusion/score models, stochastic calculus, probability measures over probability measures, sparse optimization, deep reinforcement learning, nonlinear programming, convex optimization...
Control: MPPI, path-integral control, pseudospectral transcription based numerical optimal control, structured H_∞, contraction metrics, robust/adaptive control.
Modeling: data-based dynamical modeling, Koopman/Perron–Frobenius operators, Uncertainty Quantification via Polynomial Chaos Expansions and Gaussian Mixture Models / Unscented Transformations, Sensor Fusion like Covariance Intersection, sim-to-real V&V.

SELECTED TECHNICAL RESULTS

- Built optimization-driven launch-vehicle configuration/trajectory design software with realistic UQ margins and Monte Carlo validation.
- Developed hybrid gradient-based and gradient-free optimizers in MATLAB/Octave/Python for mission/system design, nonlinear programming, and robust trajectory optimization.
- Used PCE, unscented filtering, ensemble pseudospectral optimal control, and replay/log analysis to stress-test closed-loop guidance algorithms.
- Built control-oriented model-learning pipelines using Koopman/Perron–Frobenius ideas, data-driven dynamics, and validation/verification loops.

EARLIER APPLIED PROJECTS

- ASELSAN collision early-warning/collision-avoidance systems for N to N analysis and robust and optimal maneuver design.
- ASELSAN re-entry hypersonics CFD validation: aerodynamic/heating validation context for trajectory and control studies.
- Mission/system design: Apophis microsatellite concept and expandable lunar-base concept; system-level analysis and multidisciplinary trade studies.

RESEARCH DIRECTION

- Robotics autonomy; interpretable LBMs; data-driven dynamics; robust policy learning; VLA evaluation; stochastic control; measure-valued dynamics; nonlinear programming; uncertainty-aware planning; safety V&V.

EXPERIENCE

FieldAI — Irvine, CA, USA
Research Intern, Data-Driven Dynamics / VLA and Large Behavior Models — *Summer 2026*
- Gaining practical exposure to **field foundation models, Vision-Language-Action models, large behavior models, diffusion-style policies, Linux/Git/Docker workflows, and validation/verification** in a robotics AI environment.
- Main technical angle remains **learning dynamics from data** for control/autonomy: rollout analysis, capability/failure-mode evaluation, robustness validation, and sparse/optimization-based interpretability of policies.

Georgia Institute of Technology — Atlanta, GA, USA
Graduate Research Assistant, Stochastic Optimal Control / Robotics Autonomy — *Aug. 2024–Present*
- Developing theory and computation for **stochastic optimal control, distribution/covariance steering, optimal stopping, Fokker–Planck transformations, nonlinear optimal transport, and free-final-time for distribution steering**.
- Studying **probability measures over probability measures**, measure-valued dynamics, stochastic calculus, and learning-based approximations for robust planning/control under uncertainty.
- Building MPPI/path-integral algorithms combining costate information, sparsity, optimal transport, and nonlinear programming for robot autonomy and trajectory optimization.

ROKETSAN Inc. — Istanbul, Turkiye
Senior Guidance, Navigation & Control Engineer — *Jan. 2021–Aug. 2024*
- Developed launch-vehicle **multidisciplinary optimization** and trajectory/configuration design software in MATLAB/Octave/Python using convex optimization, sparse nonlinear programming, pseudospectral optimal control with sparse Hessian approximation strategies, UQ margins, PCE/unscented filtering, Monte Carlo validation, and hybrid gradient-based/gradient-free optimizers.
- Designed and verified real-time guidance/control algorithms for orbital insertion, rendezvous/docking, re-entry, and spacecraft mission design: IGM/UPEG, Lambert solvers, robust autopilots, structured H_∞, contraction metrics.
- Built automated log-analysis/replay tools, vectorized Monte Carlo validation, unit-tested simulation infrastructure, optimal control allocation, Sinkhorn/optimal-transport modules, and Koopman/Perron–Frobenius data-driven dynamics models for control design and verification.

PROJECTS / RESEARCH PORTFOLIO

- **Path Integral Control:** Sparse MPPI for dual control inside of MPPI using sparse kernel functions and adaptive control for updating the feature weights. optimal-transport-based MPPI; costate-informed rollout for path-integral control that exploits the optimality structure of optimal control via costate dynamics; double-sided path-integral methods for gradient-free nonlinear programming for constrained optimization problems under distributional constraints; cost-informed MPPI for optimal transport - optimal control, applied to spacecraft formation reconfiguration in elliptical orbit.
- **DNN Warm-start Pseudo-spectral / Robust Policy Learning via Ensemble Optimal Control:** costate-dynamics warm starts via covector mapping theory and ensemble optimal control. Checkpoint logic to shift from robust to deterministic optimal control logic. Neural-network policies for integrated guidance/control under uncertainty, including translational and rotational policy modules.

SELECTED PUBLICATIONS / MANUSCRIPTS

- **Continuous-Time Covariance Steering with Common Free-Final Time, 2026**: theory for turnpikes used in distribution steering problems, studying phenomenons that appear in optimal policies in varying small- and large-time horizon settings
- **Nonlinear Stochastic Optimal Control and Optimal Stopping using Fokker–Planck Transformation, 2026**: theory for stochastic control, distribution steering, optimal stopping and common optimal free-final time transversality conditions, and diffusion/score-based theoretical work.
- **Stochastic Trajectory and Controller Optimization via Optimal Recovery Diffusion Control, 2024**: diffusion-control viewpoint to find the stochastic optimal recovery policies for a Mars re-entry vehicle using score matching algorithms via sparse gaussian kernel functions
- **Stochastic Trajectory and Robust Controller Optimization via Contractive Optimal Control, 2024**: integrated guidance and control architecture that designs both reference and the robust controller using control contraction metrics inside nonlinear programming
- **Stochastic Optimal Control under Non-Gaussian Uncertainties via Entropy Minimization and Dynamical Indicators, 2024**: entropy-based uncertainty treatment, nonlinear stochastic dynamics, and robustness indicators for safety-critical autonomy.
- **Real-Time Optimal Control of a 6-DOF State-Constrained Close-Proximity Rendezvous Mission, Acta Astronautica, 2023**: ensemble optimal control, warm-start Bellman pseudospectral control, DNN costate-policy learning, reachability checkpoints, and robust recursive replanning under estimation errors.
- **Robust Trajectory Optimization of Re-entry Flight with Prescribed Endpoint Region via Sparse Grid Ensemble Pseudospectral Optimal Control, Advances in Space Research, 2023**: sparse-grid uncertainty propagation, ensemble pseudospectral optimization, endpoint-region constraints, and robustness against dispersions.
- **EPOCS / Modular Multi-Phase Pseudospectral Optimal Control Solver, 2023**: software-oriented optimal-control work: phase linkage, constraints, warm-starts, solver validation, and reusable trajectory-optimization infrastructure.
- **Recovery Ensemble Control; Desensitized Ensemble Guidance; Multi-phase Robust Trajectory Optimization of a Hybrid Guidance Architecture for Launch Vehicles, 2023**: robust guidance, integrated trajectory/control design, uncertainty-aware optimization, and verification of safety-critical autonomy.
- **Integrated Guidance and Fuel-Optimal Control of a Space Servicing Vehicle via Warm-Start Bellman Pseudospectral Method, 2022**: learning robust policies through neural networks, costate/covector warm starts, modular optimal-control software, and nonlinear programming.